

Name: _____

Date: _____

AREA AND PERIMETER, (CIRCUMFERENCE OF A CIRCLE)

LO: I can calculate the circumference of a circle using $C = \pi d$ or $C = 2\pi r$.

Circumference

The **circumference** is the distance around a circle. The **diameter** goes across through the centre; the **radius** is half the diameter.

Formulae: $C = \pi d$ or $C = 2\pi r$ (same thing because $d = 2r$).
Use $\pi = 3.14159...$ or the π button on a calculator.

Quick examples

●	$d = 10$	$C = 10\pi$	31.4 cm	$C = \pi d$
●	$r = 7$	$C = 14\pi$	44.0 cm	$C = 2\pi r$
●		π	3.14159...	irrational number
●		$C = \pi r$ (not πd)		that halves the answer
●		$C = \pi r^2$		that is the area formula

Key idea

Circumference = $\pi \times$ diameter. If given the radius, double it first (or use $C = 2\pi r$).

$$C = \pi \times d$$

π is approximately 3.14

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – given diameter

Find the circumference of a circle with diameter 8 cm.

$$C = \pi d$$

$$C = \pi \times 8$$

$$C = 25.1 \text{ cm (1 d.p.)}$$

$$= 25.1 \text{ cm}$$

Multiply π by the diameter.

Example 2 – given radius

Find the circumference of a circle with radius 5 cm.

$$C = 2\pi r$$

$$C = 2 \times \pi \times 5$$

$$C = 31.4 \text{ cm (1 d.p.)}$$

$$= 31.4 \text{ cm}$$

Double the radius to get the diameter, then multiply by π .

Example 3 – finding diameter

A circle has circumference 50 cm. Find the diameter.

$$C = \pi d, \text{ so } d = C \div \pi$$

$$d = 50 \div \pi$$

$$d = 15.9 \text{ cm (1 d.p.)}$$

$$d = 15.9 \text{ cm}$$

Rearrange: divide circumference by π .

Example 4 – semicircle perimeter

Find the perimeter of a semicircle with diameter 12 cm.

$$\text{Curved part} = \frac{1}{2} \times \pi \times 12 = 6\pi = 18.85 \text{ cm}$$

$$\text{Straight edge} = 12 \text{ cm}$$

$$\text{Total perimeter} = 18.85 + 12 = 30.85 \text{ cm}$$

$$= 30.8 \text{ cm (1 d.p.)}$$

Semicircle perimeter = half circumference + diameter.

YOUR TURN – PRACTICE (deliberate practice)

1. Given diameter

Find the circumference (1 d.p.). Use $\pi = 3.14$.

- a) $d = 10 \text{ cm}$
- b) $d = 6 \text{ cm}$
- c) $d = 15 \text{ cm}$
- d) $d = 25 \text{ cm}$

2. Given radius

Find the circumference (1 d.p.). Use $\pi = 3.14$.

- a) $r = 4 \text{ cm}$
- b) $r = 7 \text{ cm}$
- c) $r = 10 \text{ cm}$
- d) $r = 3.5 \text{ cm}$

3. Finding radius or diameter

Find the missing measurement (1 d.p.).

- a) $C = 62.8 \text{ cm}$; find d
- b) $C = 44.0 \text{ cm}$; find r
- c) $C = 100 \text{ cm}$; find d
- d) $C = 75 \text{ cm}$; find r

4. Word problems

Use $\pi = 3.14$. Give answers to 1 d.p.

- a) A circular pond has diameter 4 m. Find circumference.
- b) A wheel has radius 30 cm. Distance in one turn?
- c) A plate has circumference 78.5 cm. Find diameter.
- d) Circular track radius 50 m. Distance for 1 lap?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $C = \pi d$ gives the circumference ☐

Correct answer: _____

Reason: _____

b) $C = \pi r^2$ gives the circumference ☐

Correct answer: _____

Reason: _____

c) Diameter = $2 \times$ radius ☐

Correct answer: _____

Reason: _____

d) Semicircle perimeter = half the circumference only ☐

Correct answer: _____

Reason: _____

e) π is approximately 3.14 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A bicycle wheel has radius **35 cm**. How many complete turns does it make to travel **1 km**? Show your working.

Expression: _____

Simplified: _____

b) A running track consists of two straight sections of **100 m** and two semicircular ends. The total distance around the track is **400 m**. Find the radius of each semicircular end.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

